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PENDIDIKAN MENENGAH
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TEST PROJECT
MODUL A – LINUX ENVIRONMENT

IT NETWORK SYSTEMS ADMINISTRATION

Introduction

PT Nusantara Digital Services is a growing technology company that provides web hosting and communication services for several local businesses. To improve security and reliability, the IT department is redesigning their infrastructure by separating the network into two zones: an Internal Network (INT) for critical services and a Demilitarized Zone (DMZ) for public-facing applications. A firewall server acts as the main gateway between the internet and the company networks, controlling traffic and protecting internal resources. The internal server (int-mail) provides core infrastructure services including DNS resolution, LDAP directory services for authentication, and a mail system for employee communication. In the DMZ, a reverse proxy load balancer (web-lb) distributes incoming web traffic to two backend web servers (`web-01` and `web-02`) to ensure high availability of the company's web services. Remote users access the infrastructure through the internet using a client machine that will be used to verify whether services such as web access, email delivery, and DNS resolution are functioning correctly according to the company's network policies. As the system administrator, your task is to deploy and configure all required services on the provided servers and ensure that the entire infrastructure operates securely and reliably.

Login

Username: root/user

Password: Skills39!

System Configuration

Timezone: Asia/Jakarta

General Configuration

Fully Qualified Domain Name	Ipv4	Services
fw.nusantara.id	INT: 10.0.1.254/24 DMZ: 10.0.2.254/24 WAN: 200.200.200.254/24 MGMT: 172.20.0.12/24	VPN (openvpn), firewall (nftables)
int-mail.nusantara.id	INT: 10.0.1.10/24 MGMT: 172.20.0.11/24	DNS Server (bind9), LDAP (slapd), CA (openssl), Mail Server (Postfix + Dovecot), WebMail (Roundcube), ansible
web-lb.nusantara.id	DMZ: 10.0.2.10/24 MGMT: 172.20.0.13/24	HAProxy
web-01.nusantara.id	DMZ: 10.0.2.11/24 MGMT: 172.20.0.14/24	Web Server (Nginx)
web-02.nusantara.id	DMZ: 10.0.2.12/24 MGMT: 172.20.0.15/24	Web Server (Nginx)
client.nusantara.id	WAN: 200.200.200.100/24 MGMT: 172.20.0.16/24	E-mail client, VPN client

Networking

To make your life a bit easier, networking has been preconfigured on all machines.

Notes

For the assessment process, we use a system that connects to every server through the management network (MGMT) using TCP port 9000.

Do not block inbound TCP port 9000 on any server. If you configure host-based firewall rules, ensure that access on TCP port 9000 remains allowed on the MGMT interface/network.

Part 1 INT (Internal)

int-mail.nusantara.id

DNS

Set up a DNS server using BIND9 to manage domain names

1. Install & Configure BIND9
2. Create DNS Zones
    - Create a forward zone for: nusantara.id
    - Create a reverse zone for the INT network: 1.0.10.in-addr.arpa
    - Create a reverse zone for the DMZ network: 2.0.10.in-addr.arpa
    - Ensure all zones are authoritative and answer SOA queries correctly
3. Add DNS Records
    - Add forward records for the hosts listed in the table below.
    - Add reverse records only for the IP addresses listed in the PTR table below.
    - Configure at least one valid NS record for nusantara.id.
    - Add additional records on the tables below.

Type	Name	Value/Target
A	www	200.200.200.254
A	int-mail	10.0.1.10
A	web-lb	10.0.2.10
A	web-01	10.0.2.11
A	web-02	10.0.2.12
A	ansible	10.0.2.12
NS	@	int-mail.nusantara.id
CNAME	ldap	int-mail.nusantara.id
CNAME	www.int	web-lb.nusantara.id
MX	@	10 int-mail.nusantara.id

Type	IP Address	Value/Target
PTR	10.0.1.10	int-mail.nusantara.id
PTR	10.0.2.10	web-lb.nusantara.id
PTR	10.0.2.11	web-01.nusantara.id
PTR	10.0.2.12	web-02.nusantara.id

LDAP

Install and configure LDAP using slapd to manage centralized user accounts. These users will be used for authentication on the mail server.

1. Install LDAP using slapd and ldap-utils
2. Configure the LDAP directory structure
    - Use the following Base DN: dc=nusantara,dc=id
    - Create the following Organizational Units:
        - ou=users,dc=nusantara,dc=id
        - ou=groups,dc=nusantara,dc=id
        - ou=services,dc=nusantara,dc=id
3. Create LDAP Groups
    - Create the following groups under ou=groups,dc=nusantara,dc=id:
        - mail
4. Create LDAP Users
    - Create the following users under ou=users,dc=nusantara,dc=id
    - Assign users into groups based on the table below
    - These users will be used for service authentication

Full Name	Username (uid)	Password	Groups	Email Address
Boaz Salossa	boaz	Skill39!	mail	boaz.salossa@nusantara.id
Budi Sudarsono	budi	Skill39!	mail	budi.sudarsono@nusantara.id

5. Create service bind accounts
    - Create the following service accounts under ou=services,dc=nusantara,dc=id
    - These accounts will be used by services to search and authenticate users in LDAP

Full Name	Username (uid)	Password	Groups
Mail Server	uid=svc-mail,ou=services,dc=nusantara,dc=id	Skill39!	Used by Postfix/Dovecot to search and authenticate mail users

CA

Use OpenSSL to create your own Certificate Authority (CA) and generate server certificates for web and mail services.

1. Create Root CA
   - Create one Root CA with Common Name (CN): Nusantara Root CA
   - Other certificate subject fields such as Country, State, or Organization may use any value
   - The Root CA certificate must include the following extensions:
     X509v3 Basic Constraints: critical
       CA:TRUE
     X509v3 Key Usage: critical
       Certificate Sign, CRL Sign
2. Create Certificates for Servers
   - Generate and sign the following certificates directly using the Root CA
   - Use Subject Alternative Name (SAN) on each certificate
   - The web certificate must be valid for:
     i. www.nusantara.id
     ii. www.int.nusantara.id
   - The mail certificate must be valid for:
     i. int-mail.nusantara.id
3. Validation Requirements
   - Each server certificate must be verifiable using the Root CA
   - The web certificate must contain the required SAN entries
   - The mail certificate must contain the required SAN entry
   - Install the Root CA certificate into the operating system trust store on all hosts listed in the General Configuration
   - After installation, the Root CA must be recognized by the operating system as a trusted Certificate Authority
4. Save All Certificates
   - Save the certificates in this folder: /opt/grading/ca
   - Use the following filenames:
     i. ca.pem: Root CA certificate
     ii. web.pem: Web server certificate
     iii. mail.pem: Mail server certificate

Mail Server

Set up a mail server using Postfix and Dovecot so LDAP users can send and receive emails for the domain nusantara.id.

1. Install Mail Server Software
   - Install Postfix as the SMTP server
   - Install Dovecot as the IMAP server
   - Install Roundcube as the webmail client
2. Mail Server Scope
   - The mail server only needs to handle email for the local domain: nusantara.id
   - Mailboxes must use virtual mailbox storage with Maildir format
   - Users must be able to read email using IMAP
   - Users must be able to send email using authenticated SMTP submission
3. TLS Requirements
   - Enable TLS for IMAP (993) and SMTP submission (587)
   - Use the mail certificate created in the Certificate Authority (OpenSSL) section
   - The certificate presented by the mail service must be valid for int-mail.nusantara.id
   - Clients that trust Nusantara Root CA must be able to verify the certificate
4. LDAP Integration
   - Use LDAP as the authentication source for mail users
   - Only users in the mail group may authenticate to the mail service
   - Users must log in using their uid
   - The email address for each user must be taken from the mail attribute in LDAP
   - Authentication must come from LDAP, but mailbox storage may remain local on the mail server
   - Prepare working virtual mailboxes for at least these users:
     - boaz
     - budi
5. Functional Testing Requirements
   - User boaz must be able to:
     - Log in to the mail service using his LDAP credentials
     - Access his mailbox using IMAP
     - Send an email to boaz.salossa@nusantara.id
     - Receive and read that email in his own mailbox
6. Roundcube Requirements
   - Configure Roundcube to connect to the local Postfix and Dovecot services
   - Roundcube must be accessible at: https://int-mail.nusantara.id
   - Clients that trust Nusantara Root CA must be able to access https://int-mail.nusantara.id without certificate warnings
   - User boaz must be able to log in to Roundcube using his mail credentials
   - After login, boaz must be able to view his inbox

Ansible

Use int-mail.nusantara.id as the Ansible control node to automate configuration on web-02.nusantara.id.

1. Ansible Control Node
   - Install and configure Ansible
   - Configure SSH access from int-mail.nusantara.id to web-02.nusantara.id
   - In this section, web-02.nusantara.id is the only managed node
2. Inventory and Playbook Location
   - Save the inventory file in: /opt/ansible/inventory.ini
   - Save all playbooks in: /opt/ansible/
3. Playbook 1: Create Users use name 01-users.yaml
   - Create one playbook to create the following local users on web-02.nusantara.id:
     - user1
     - user2
     - ....
     - user10
   - Use password Skill39!
   - Each user must have a home directory
   - The playbook must be safe to run multiple times without failing
4. Playbook 2: Configure Nginx Virtual Host use name 02-web.yaml
   - Create one playbook to install and configure Nginx on web-02.nusantara.id
   - Create a virtual host for: ansible.nusantara.id
   - The virtual host must serve a simple page that clearly identifies the site, for example:
     - Hello from ansible.nusantara.id
   - Ensure the nginx service is enabled and running
   - The playbook must be safe to run multiple times without failing

Part 2 Web

web-lb.nusantara.id, web-01.nusantara.id & web-02.nusantara.id

Web Service (HAProxy and Nginx)

Set up a simple web service topology using one HAProxy load balancer and two Nginx backend servers.

1. Topology
   - web-lb.nusantara.id must run HAProxy
   - web-01.nusantara.id must run Nginx on port 8080
   - web-02.nusantara.id must run Nginx on port 8080
   - HAProxy on web-lb.nusantara.id must distribute traffic to:
     - 10.0.2.11:8080
     - 10.0.2.12:8080
2. Web Server Requirements
   - On web-01 and web-02, configure Nginx to listen on port 8080
   - Each backend server must serve a different page so load balancing can be observed easily
   - The page content must include the hostname of the backend server
   - A simple example is:
     - Hello from web-01
     - Hello from web-02
3. Load Balancer Requirements
   - HAProxy must accept HTTP and HTTPS traffic on web-lb.nusantara.id
   - All HTTP requests must be redirected to HTTPS
   - HAProxy must perform TLS termination and forward requests to the backend Nginx servers using HTTP
   - Add an HTTP response header via-proxy with the value web-lb.nusantara.id
4. TLS Requirements
   - Use the web certificate created in the Certificate Authority (OpenSSL) section
   - The certificate presented by HAProxy must be valid for:
     - www.nusantara.id
     - www.int.nusantara.id
   - A client that trusts Nusantara Root CA must be able to validate the certificate chain

Part 3 Firewall

fw.nusantara.id

nftables

Create firewall and NAT rules using nftables to control which traffic is allowed or blocked in your network.

1. Default Policy
    - Apply a default deny policy for forwarded traffic
    - Any traffic that is not explicitly allowed in this section must be blocked
2. Allow Internet Access
    - Hosts in the INT network (10.0.1.0/24) must be able to access the internet through the firewall
    - Hosts in the DMZ network (10.0.2.0/24) must be able to access the internet through the firewall
3. NAT Masquerade
    - When traffic from the INT or DMZ networks goes out through the WAN, the firewall must perform source NAT using masquerade
4. Port Forwarding
    - Forward incoming traffic from the WAN to the following internal services:

Port	Protocol	Forward To
80	TCP	10.0.2.10
443	TCP	10.0.2.10
53	TCP/UDP	10.0.1.10

5. VPN Access
    - VPN clients connected to fw.nusantara.id must be able to reach both:
        - 10.0.1.0/24 (INT)
        - 10.0.2.0/24 (DMZ)
6. dont block tcp port 9000 for all

VPN (openvpn)

Set up a remote-access VPN server using OpenVPN on fw.nusantara.id.

1. OpenVPN Server
    - Install and configure OpenVPN
    - Use tun mode
    - Use UDP port 1194
    - Use the VPN network: 10.0.10.0/24
2. Certificate Requirements
    - Use the Certificate Authority created in the Certificate Authority (OpenSSL) section
    - The OpenVPN server must use a server certificate signed by your Root CA
    - A VPN client that trusts Nusantara Root CA must be able to verify the VPN server certificate
3. User Authentication
    - Use system users for VPN authentication
    - Create a system user boaz on fw.nusantara.id for VPN authentication
    - User boaz must be able to authenticate for OpenVPN using system credentials
4. Access Requirements
    - A connected VPN client must be able to access:
        - 10.0.1.0/24 (INT)
        - 10.0.2.0/24 (DMZ)

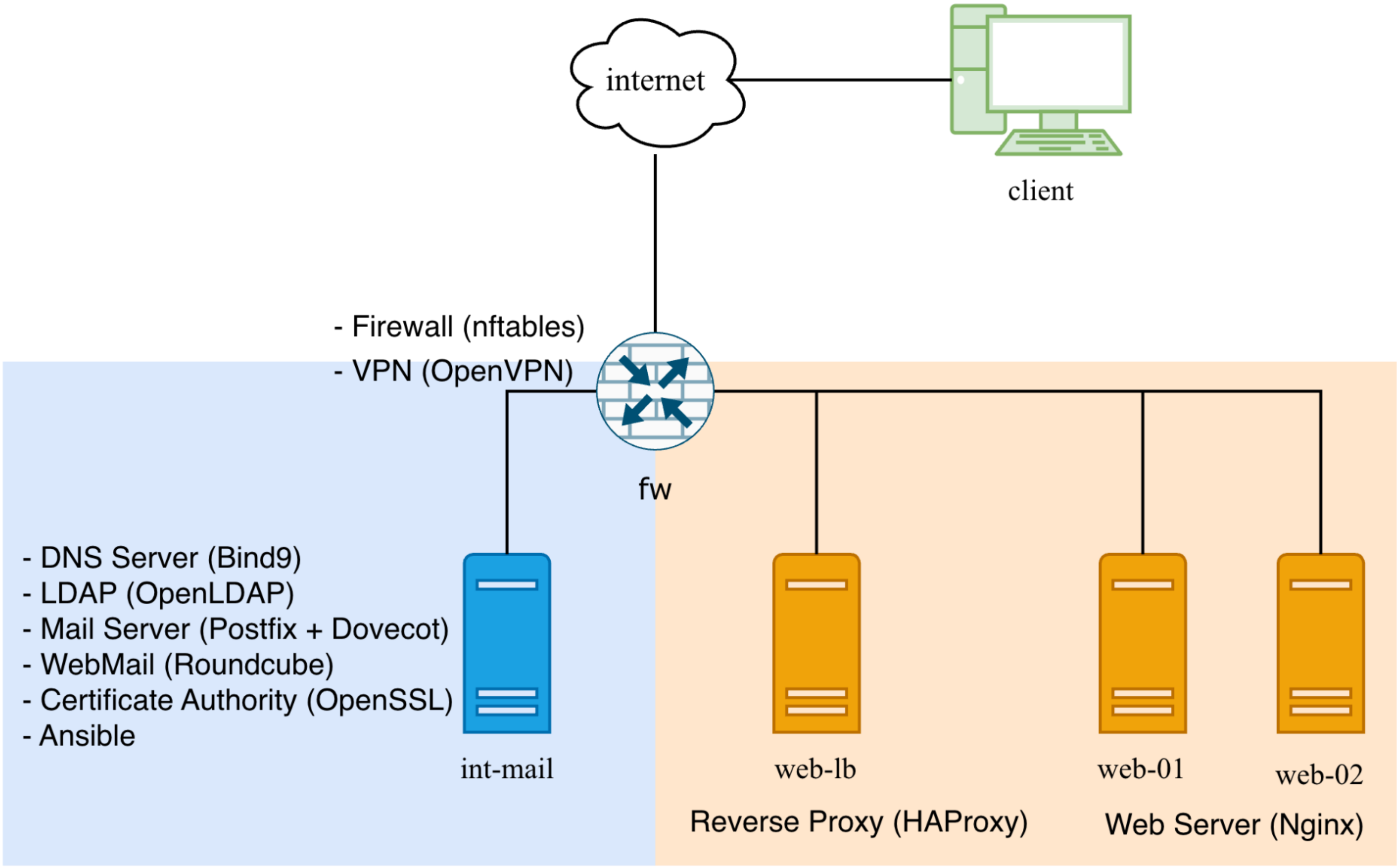
Part 4 Client

The client machine client.nusantara.id represents a remote workstation that connects securely to internal and DMZ services through VPN. It is used to validate the overall infrastructure by testing access to web and mail services.

General Configuration

1. Create a new local user boaz (Boaz Salossa) with the password Skill39!
    - Configure user boaz so it can run sudo without password
2. Configure OpenVPN client to connect to fw.nusantara.id
    - Use the OpenVPN profile provided for user boaz
    - Save the VPN client configuration as:
      /etc/openvpn/client/client.ovpn
    - Save the OpenVPN username and password in:
      /etc/openvpn/auth/user.txt
    - Start the VPN connection using:
      sudo openvpn --config /etc/openvpn/client/client.ovpn
    - Ensure the client can resolve internal DNS records required for validation after the VPN connection is established
3. Root CA Trust
    - Install Nusantara Root CA into the operating system trust store on the client machine
    - The client must trust certificates issued by this Root CA
4. Validation
    - The client must be able to access https://www.nusantara.id through the internet without certificate errors
    - After connecting to the VPN, the client must be able to access https://www.int.nusantara.id without certificate errors
    - After connecting to the VPN, the client must be able to access Roundcube webmail at https://int-mail.nusantara.id without certificate errors
    - After connecting to the VPN, the client must be able to access internal services in the INT and DMZ networks according to the infrastructure that has been configured

Topology


internet
client
- Firewall (nftables)
- VPN (OpenVPN)
fw
- DNS Server (Bind9)
- LDAP (OpenLDAP)
- Mail Server (Postfix + Dovecot)
- WebMail (Roundcube)
- Certificate Authority (OpenSSL)
- Ansible
int-mail
web-lb
web-01
web-02
Reverse Proxy (HAProxy)
Web Server (Nginx)